

Practical 6

17ce049

Tips

Introduction:

This exercise was created based on the tutorial and documentation from [Seaborn](https://stanford.edu/~mwaskom/software/seaborn/index.html) (<https://stanford.edu/~mwaskom/software/seaborn/index.html>).

The dataset being used is tips from Seaborn.

Step 1. Import the necessary libraries:

In [18]:

```
import pandas as pd

# visualization libraries
import matplotlib.pyplot as plt
import seaborn as sns

# print the graphs in the notebook
% matplotlib inline

# set seaborn style to white
sns.set_style("white")
```

Step 2. Import the dataset from this [address](https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/07_Visualization)

(https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/07_Visualization)

Step 3. Assign it to a variable called tips

In [10]:

```
url = 'https://raw.githubusercontent.com/guipsamora/pandas_exercises/master/07_Visualization/07_Visualization/tips.csv'
tips = pd.read_csv(url)

tips.head()
```

Out[10]:

	Unnamed: 0	total_bill	tip	sex	smoker	day	time	size
0	0	16.99	1.01	Female	No	Sun	Dinner	2
1	1	10.34	1.66	Male	No	Sun	Dinner	3
2	2	21.01	3.50	Male	No	Sun	Dinner	3
3	3	23.68	3.31	Male	No	Sun	Dinner	2
4	4	24.59	3.61	Female	No	Sun	Dinner	4

Step 4. Delete the Unnamed 0 column

In [12]:

```
del tips['Unnamed: 0']

tips.head()
```

Out[12]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

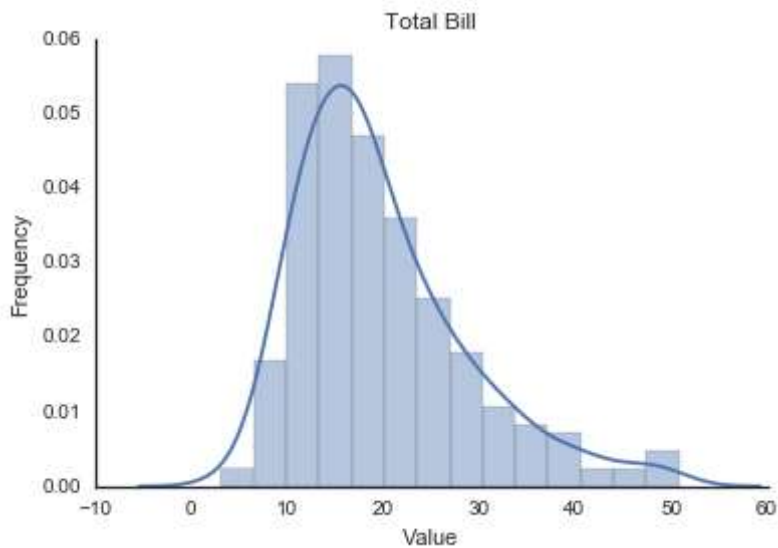
Step 5. Plot the total_bill column histogram

In [37]:

```
# create histogram
ttbill = sns.distplot(tips.total_bill);

# set labels and titles
ttbill.set(xlabel = 'Value', ylabel = 'Frequency', title = "Total Bill")

# take out the right and upper borders
sns.despine()
```



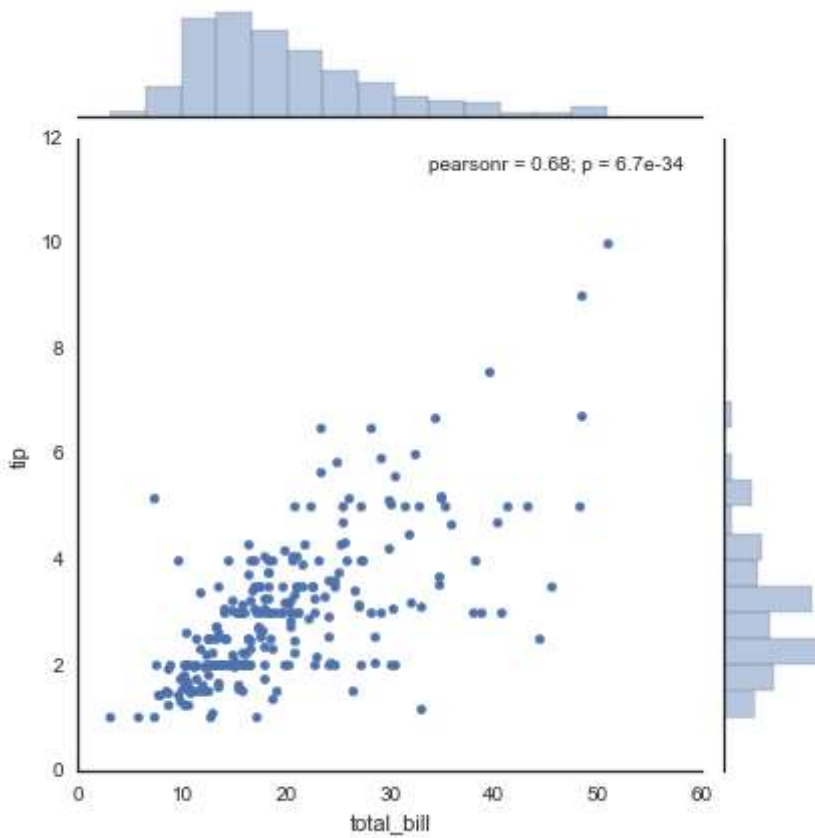
Step 6. Create a scatter plot presenting the relationship between total_bill and tip

In [46]:

```
sns.jointplot(x="total_bill", y="tip", data=tips)
```

Out[46]:

<seaborn.axisgrid.JointGrid at 0x1197d84d0>



Step 7. Create one image with the relationship of total_bill, tip and size.

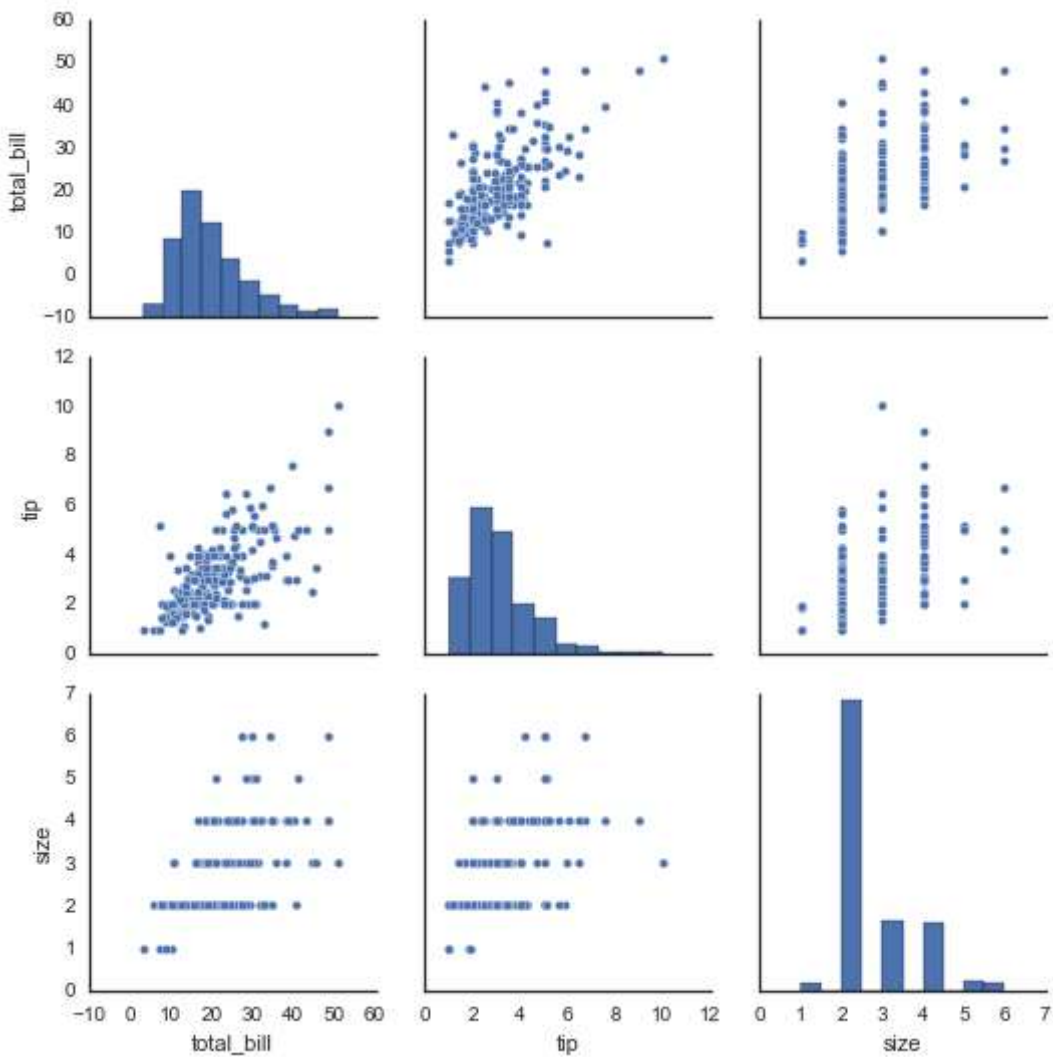
Hint: It is just one function.

In [44]:

```
sns.pairplot(tips)
```

Out[44]:

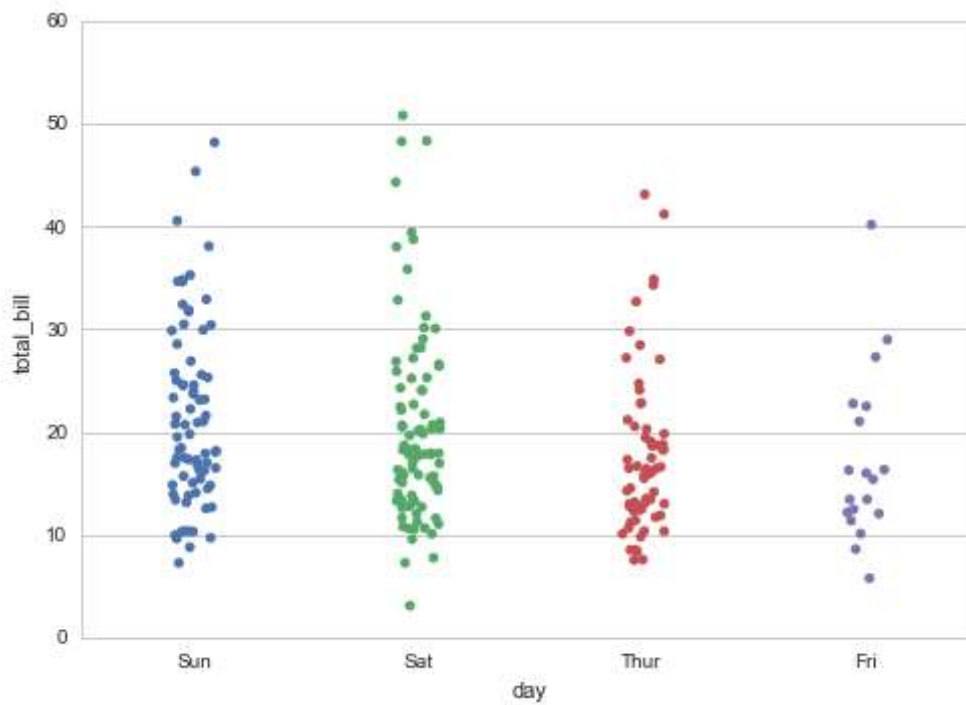
<seaborn.axisgrid.PairGrid at 0x11844c090>



Step 8. Present the relationship between days and total_bill value

In [51]:

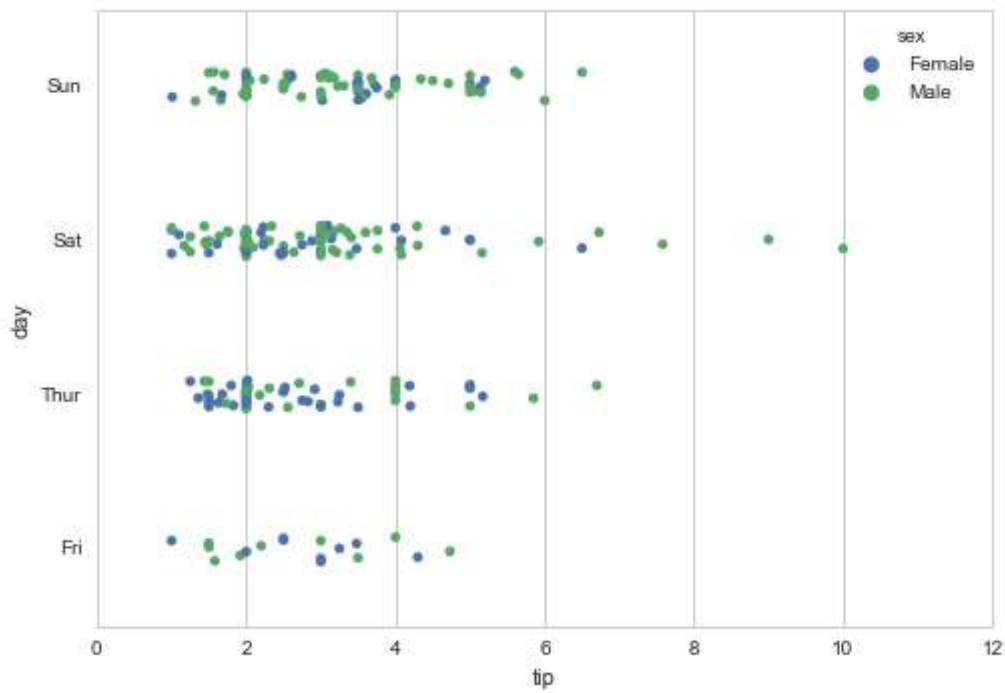
```
sns.stripplot(x = "day", y = "total_bill", data = tips, jitter = True);
```



Step 9. Create a scatter plot with the day as the y-axis and tip as the x-axis, differ the dots by sex

In [61]:

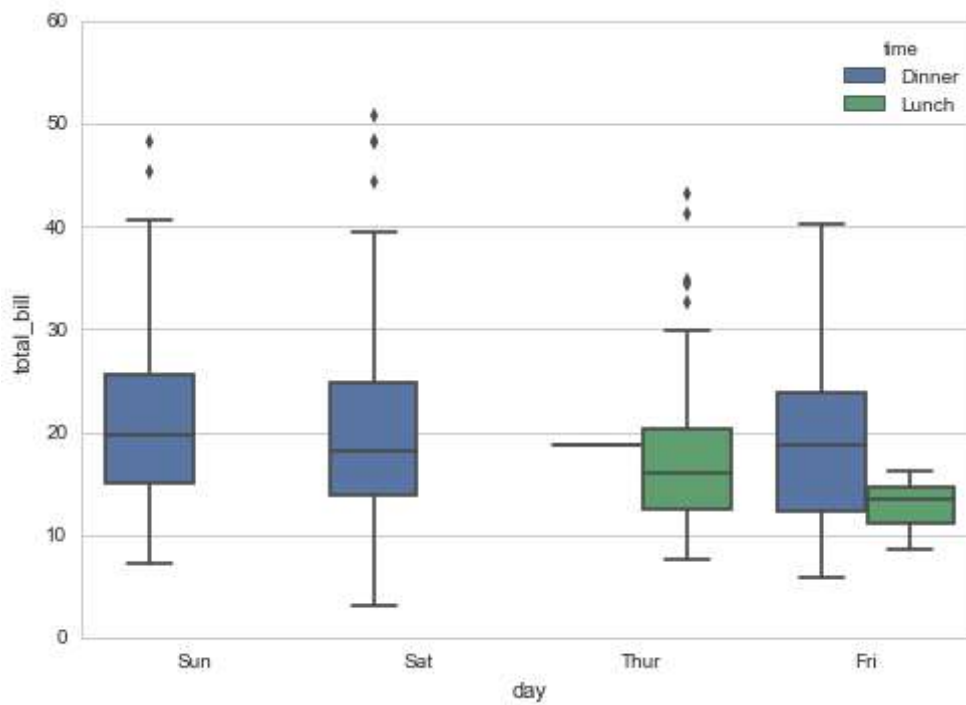
```
sns.stripplot(x = "tip", y = "day", hue = "sex", data = tips, jitter = True);
```



Step 10. Create a box plot presenting the total_bill per day differentiation the time (Dinner or Lunch)

In [58]:

```
sns.boxplot(x = "day", y = "total_bill", hue = "time", data = tips);
```

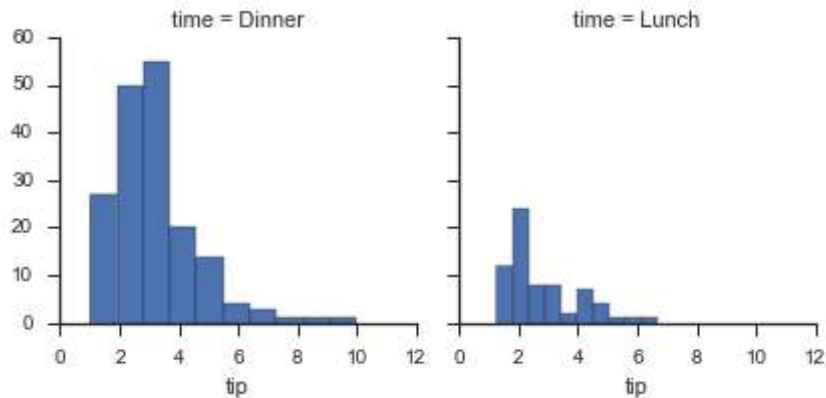


Step 11. Create two histograms of the tip value based for Dinner and Lunch. They must be side by side.

In [63]:

```
# better seaborn style
sns.set(style = "ticks")

# creates FacetGrid
g = sns.FacetGrid(tips, col = "time")
g.map(plt.hist, "tip");
```



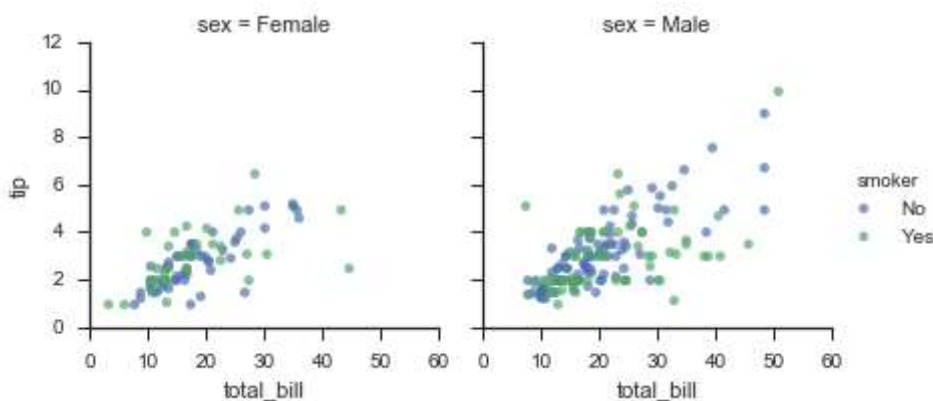
Step 12. Create two scatterplots graphs, one for Male and another for Female, presenting the total_bill value and tip relationship, differing by smoker or no smoker

They must be side by side.

In [65]:

```
g = sns.FacetGrid(tips, col = "sex", hue = "smoker")
g.map(plt.scatter, "total_bill", "tip", alpha =.7)

g.add_legend();
```



BONUS: Create your own question and answer it using a graph.

In []:

